

Quiz - 3

① $u^T u = 1$

$$H = I - 2uu^T$$

~~a)~~ $H^2 = (I \cdot I + 4uu^T \overset{1}{u}u^T - 4uu^T)$

$$H^2 = I$$

$$\Rightarrow H = H^T$$

$$H^T H = I$$

H is orthogonal, as well.

~~b)~~ $(I - 2uu^T)u = \lambda u$

$$u - 2u = \lambda u$$

$$\underline{\underline{-1 = \lambda}}$$

~~u~~

$$v^T u = 0$$

$$u^T v = 0$$

$$(I - 2uu^T)v = \lambda v$$

$$v - 2u(u^T v) = \lambda v$$

$$\boxed{\lambda = 1}$$

→ There would be $n-1$ independent $\perp u$ vectors so, $\lambda = 1$ would be repeated $n-1$ times.

→ H is diagonalizable since it is possible to find $(n-1)$ independent vectors perpendicular to u .

~~$\sum H_{ii}$~~

$$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad [u_1 \quad \dots \quad -]$$

$$uu^T = \begin{bmatrix} u_1^2 & & & \\ & u_2^2 & & \\ & & \ddots & \\ & & & u_n^2 \end{bmatrix}$$

$$H = \begin{bmatrix} 1 - 2u_1^2 & & & \\ & 1 - 2u_2^2 & & \\ & & \ddots & \\ & & & \end{bmatrix}$$

$$\sum \lambda = n - 2$$

$$\sum H_{ii} = n - 2(\underbrace{u_1^2 + u_2^2}_{1} \dots) = \underline{\underline{n-2}}$$

② $A \rightarrow$ positive definite symmetric

~~st~~ A^{-1} is symmetric as its eigenvalues are the reciprocals of λ 's of A

And since all values λ of A are positive, $1/\lambda$ would be positive as well.

~~st~~ $Q_{n \times n} \rightarrow$ orthogonal

$A \rightarrow$ positive definite

$$x A x^T > 0 \quad \forall x$$

$$B = Q A Q^{-1}$$

Hence A and B are similar and have same eigenvalues.
So, B is also positive definite.

→ partially correct (pivot test)

(c)

$$B = \begin{bmatrix} A & A \\ A & A \end{bmatrix}$$

We can see that B has dependent rows and hence it is a singular matrix.

→ Singular matrices have $\lambda = 0$
so, the matrix would be positive semidefinite.

the λ 's = # the pivots.

$$(3) \quad A = \begin{bmatrix} 0 & -1 \\ 4 & 0 \end{bmatrix}$$

$$\cancel{u)} \quad \lambda^2 + 4 = 0$$

$$\underline{\underline{\lambda = \pm 2i}} \quad \checkmark$$

$$\frac{i}{i \cdot i} = -i$$

$$\lambda = 2i$$

$$\begin{bmatrix} i/2 \\ 1 \end{bmatrix}$$

$$\lambda = -2i \quad \begin{bmatrix} -i/2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2i & -1 \\ 4 & 2i \end{bmatrix}$$

$$2i - 1 = 0$$

$$x = -\frac{i}{2}$$

$$u_0 = -2i x_1 + 2i x_2$$

$$\cancel{u)} \quad \frac{du}{dt} = Au$$

$$u_t = c_1 e^{2it} x_1 + c_2 e^{-2it} x_2$$

$$c_1 = -2i$$

$$c_2 = 2i$$

$$\cancel{A}^T A = \begin{bmatrix} 0 & -1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 4 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 0 \\ 0 & -4 \end{bmatrix} = -4 I \rightarrow A^2 \neq \underline{\underline{A^T A}}$$

$$A^T A = V \Sigma^2 V^T$$

$$\boxed{V = I} \checkmark$$

$$\begin{bmatrix} 0 & -1 \\ 4 & 0 \end{bmatrix} = U \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & -1 \\ 4 & 0 \end{bmatrix} \begin{bmatrix} -i/2 & 0 \\ 0 & i/2 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & -i/2 \\ -2i & 0 \end{bmatrix} = -i \begin{bmatrix} 0 & 1/2 \\ 2 & 0 \end{bmatrix}$$